

First integrals / Invariants

$$u'(t) = f(u)$$

$$\frac{d}{dt} I(u) = \sum_j \frac{\partial I}{\partial u_j} \frac{du_j}{dt} = I'(u) f(u)$$

so $I(u)$ is conserved if

$$I'(u) f(u) = 0 \quad \forall u.$$

Examples

harmonic oscillator

$$u' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} u \quad I(u) = \frac{1}{2} \|u\|^2 = \frac{u_1^2 + u_2^2}{2}$$

$$I'(u) = \begin{bmatrix} u_1 & u_2 \end{bmatrix} \quad f(u) = \begin{bmatrix} -u_2 \\ u_1 \end{bmatrix}$$

Lotka-Volterra or Kepler (HLW)

$$u'(t) = (v-2)u$$

$$I(u,v) = \log u - u + 2 \log v - v$$

$$v'(t) = (1-u)v$$

$$\frac{\partial I}{\partial u} = \frac{1}{u} - 1$$

$$\nabla I \cdot f = \frac{1-u}{u} (v-2)u + \frac{2-v}{v} (1-u)v$$

$$\frac{\partial I}{\partial v} = \frac{2}{v} - 1$$

$$= 0.$$

Linear invariants

$$I(u) = d^T u \Rightarrow I'(u) = d^T$$

$$\text{assume } d^T f(u) = 0 \quad \forall u$$

All RK methods preserve linear invariants,

Since

$$u_{n+1} = u_n + h \sum_j b_j f(y_j)$$

$$\Rightarrow d^T u_{n+1} = d^T u_n + h \sum_j b_j \cancel{d^T f(y_j)} \rightarrow 0$$

In fact, general linear methods preserve linear invariants.

Quadratic invariants

Let

$$u'(t) = L(u)u \quad \text{with } L^T = -L.$$

Then

$$\begin{aligned} \frac{d}{dt} \left(\frac{1}{2} u^T u \right) &= \sum_j u_j (L(u)u)_j = u^T L u \\ &= \frac{1}{2} u^T L u + \frac{1}{2} (u^T L u)^T = \frac{1}{2} (u^T L u + u^T L^T u) = 0. \end{aligned}$$

So $\frac{1}{2} \|u\|^2$ is conserved.

More generally, a quadratic invariant takes the form

$$Q(u) = u^T C u \quad \text{where } C = C^T$$

$$\text{and } u^T C f(u) = 0$$

Applying a RK method:

$$u_1 = u_0 + h \sum_j b_j f(y_j)$$

So

$$u_1^T C u_1 = u_0^T C u_0 + h \sum_i b_i u_0^T C f(y_i)$$

$$+ h \sum_j b_j f(y_j)^T C u_0 + h^2 \sum_i \sum_j b_i b_j f(y_i)^T C f(y_j)$$

Now $y_i = u_0 + h \sum_j a_{ij} f(y_j)$

Substitute this
in _____

$$\Rightarrow u_0 = y_i - h \sum_j a_{ij} f(y_j)$$

$$\Rightarrow u_1^T C u_1 - u_0^T C u_0 = 2h \sum_j b_j y_j^T C f(y_j)$$

← vanishes since
Q is
conserved

$$+ h^2 \sum_i \sum_j (f(y_i)^T C f(y_j)) (b_i b_j - b_i a_{ij} - b_j a_{ji})$$

If this vanishes,
we get conservation.

Thm. (Cooper 1987):

If the coefficients of an RK method satisfy

$$b_i a_{ij} + b_j a_{ji} = b_i b_j \quad \forall (i, j)$$

then it conserves quadratic invariants.

The implicit midpoint method

$$\begin{array}{c|c} \frac{1}{2} & \frac{1}{2} \\ \hline & 1 \end{array}$$

satisfies this, as do the Gauss methods.

They are fully implicit.

PDE invariants

For PDEs, whether a given quantity is conserved also depends on the boundary conditions. To avoid this complication, we assume periodic conditions or an unbounded domain and localized data, so boundary terms cancel or vanish.

Example: Korteweg-de Vries (KdV)

$$u_t + uu_x + u_{xxx} = 0$$

We have

$$\frac{d}{dt} \int u dx = \int u_t dx = - \int (uu_x + u_{xxx}) dx$$

$$= - \int \left(\frac{1}{2} u^2 + u_{xx} \right)_x dx = 0$$

and

$$\frac{d}{dt} \int \frac{1}{2} u^2 dx = \int u u_t dx$$

$$= - \int u (u u_x + u_{xxx}) dx$$

$$= - \int (u^2 u_x + u_{xxx}) dx = - \int \left(\frac{1}{3} u^3 + u_{xx} \right)_x dx$$

$$= 0.$$

In general, PDE conservation laws take the form

$$\frac{\partial}{\partial t} \int I(u) dx + \frac{\partial}{\partial x} \int F(u) dx = 0.$$